HDOC DAY-65

Recursive Program: Tower of Hanoi

Aim

To write a recursive C program that solves the Tower of Hanoi problem for **n disks** and displays every move required to transfer the disks from the source rod to the destination rod.

Problem Statement

There are three rods: Source (A), Auxiliary (B), and Destination (C). Initially, all disks are placed on rod A in decreasing order of size, with the largest disk at the bottom. Move all disks to rod C while following these rules:

1. Only one disk can be moved at a time.

2. Only the top disk of a rod can be moved.

3. A larger disk cannot be placed on a smaller disk.

Algorithm

1. Read the number of disks n.

2. If n equals 1, print the move from the source rod to the destination rod and return.

3. Recursively move n – 1 disks from the source rod to the auxiliary rod, using the destination rod.

4. Print the move of the largest disk from the source rod to the destination rod.

5. Recursively move the n – 1 disks from the auxiliary rod to the destination rod, using the source rod.

6. Stop when all moves have been printed.

C Program

```
#include <stdio.h>

void towerOfHanoi(int n, char source, char auxiliary, char destination)
{
    if (n == 1)
    {
        printf("Move disk 1 from %c to %c\n", source, destination);
        return;
    }

    towerOfHanoi(n - 1, source, destination, auxiliary);

    printf("Move disk %d from %c to %c\n",
           n, source, destination);

    towerOfHanoi(n - 1, auxiliary, source, destination);
}

int main(void)
{
    int n;

    printf("Enter the number of disks: ");
    if (scanf("%d", &n) != 1 || n < 1)
```

```
    {
        printf("Please enter a positive integer.\n");
        return 1;
    }

    towerOfHanoi(n, 'A', 'B', 'C');

    return 0;
}
```

Sample Output

```
Enter the number of disks: 3
Move disk 1 from A to C
Move disk 2 from A to B
Move disk 1 from C to B
Move disk 3 from A to C
Move disk 1 from B to A
Move disk 2 from B to C
Move disk 1 from A to C
```

Explanation

The function **towerOfHanoi()** calls itself with one fewer disk, so this is an example of recursion. First, it moves the smaller disks out of the way, then moves the largest disk to its destination, and finally moves the smaller disks onto it. The base case is **n == 1**, which prints one move and stops that recursive call.

Result

The recursive C program successfully solves the Tower of Hanoi problem and displays all moves needed to transfer n disks from the source rod to the destination rod.

Note: The minimum number of moves for n disks is $2^n - 1$. For 3 disks, the program prints 7 moves.